Principles of dendrochronology

Lesson 3 · 90 minutes · Verifying ring dates through shared patterns

Learning objectives

Explain shared growth patterns

Relate limiting factors to replication.

Crossdate rings

Match several features and investigate possible ring anomalies.

Support a dating decision

Distinguish calendar anchoring, overlap length and independent checks.

Review: boundaries and dates

Tree-ring boundary

Identify an annual increment from anatomical changes.

False and discontinuous rings

Counting one radius can introduce extra or missing years.

Question for today

What can comparison with other trees establish?

Counting and crossdating

Approach	What it establishes
Ring counting	Counts visible increments; missed or extra boundaries can shift the result.
Crossdating	Tests ring dates by matching shared patterns among series.
Calendar anchoring	Connects the sequence to a securely dated reference.

Factors that limit growth

Growth

Limiting factor

Water availability can limit growth at dry sites.

Growing-season temperature can limit growth at cold sites.

The relevant factor varies with species, site and period.

Choosing sites and trees for the question

Climate questions

Select sites where trees respond to the climate variable of interest.

Forest-history questions

Look for trees and spatial patterns recording competition or disturbance.

Conditions for comparison

The same species or region does not guarantee matching growth.

Several influences contribute to ring width

Age and stem size

Shared environmental variation

Competition, disturbance and individual effects

→ **Ring-width variation**

Conceptual influences; independence and simple additivity are not assumed.

Uniformitarianism and testing growth responses

Uniformitarianism

Use present processes to help interpret past records.

Responses may change

Age, site and climate changes can alter growth responses.

Reconstruction needs further tests

Verify dating and environmental interpretation separately.

Replication across radii, trees and sites

Multiple radii of one tree **Different trees** **Different sites**

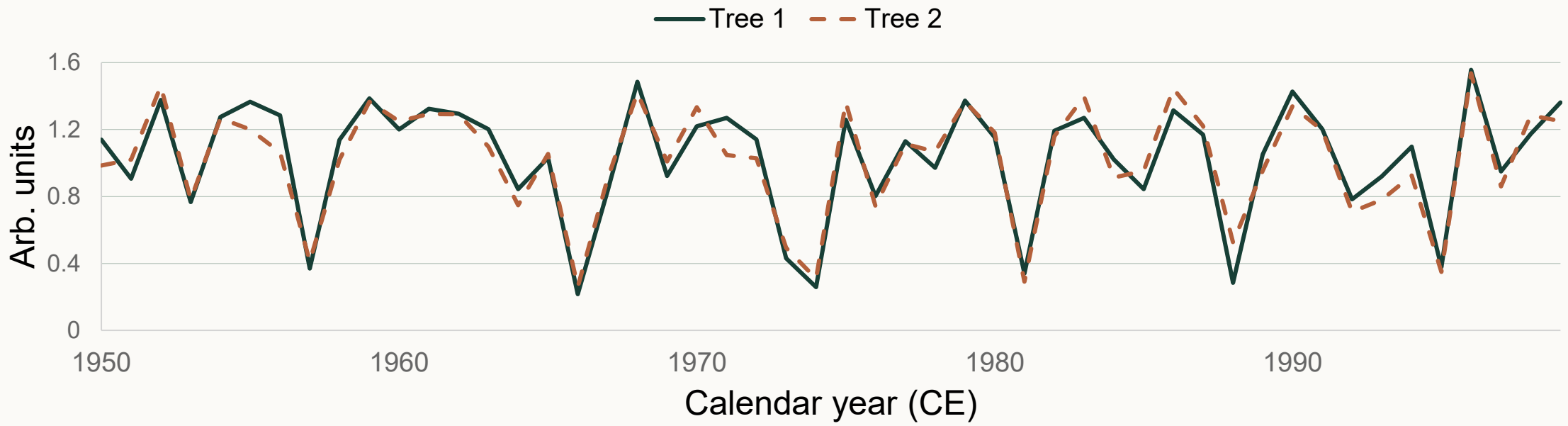
Scope	Example	What it checks
Radii	Tree 1, radii a / b	Local absences or reading errors
Trees	Tree 1 and tree 2	Shared patterns between individuals
Sites	Sites X and Y	Spatial extent of shared responses

Four samples need not represent four trees.

The basis of crossdating

Compare the sequence of variation in different trees.

Absolute widths can differ while shared features remain.



Synthetic teaching data / schematic example; not empirical measurements.

From one marker year to a sequence

A single narrow-ring match is insufficient.

Check surrounding variation, spacing and the whole overlap.



Example: 1966 → 1973 → 1981 → 1988

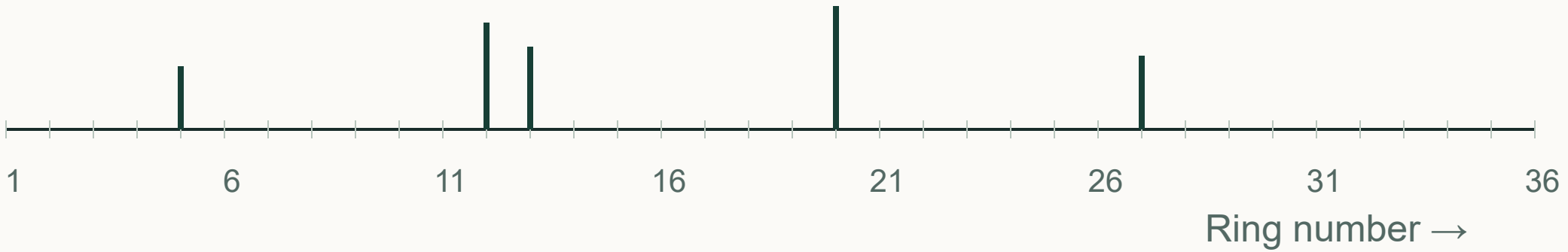
Synthetic teaching data / schematic example; not empirical measurements.

Skeleton plots

Narrow rings receive longer marks relative to neighbours.

Normal rings can remain unmarked; compare feature spacing.

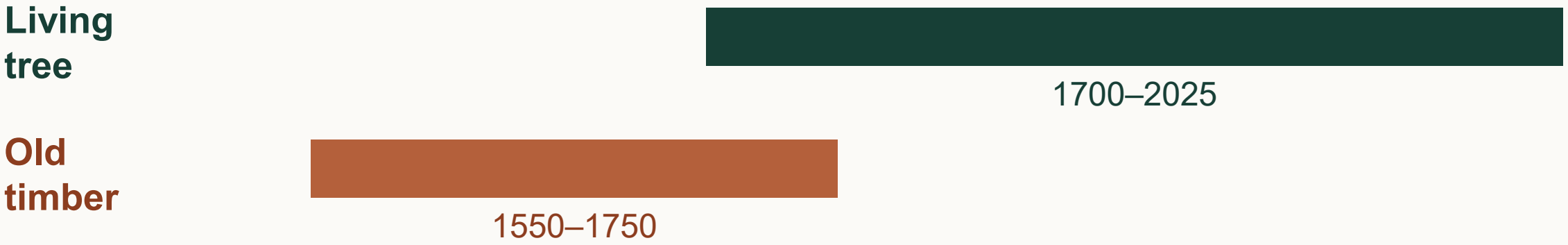
Marks represent relative features, not measured widths.



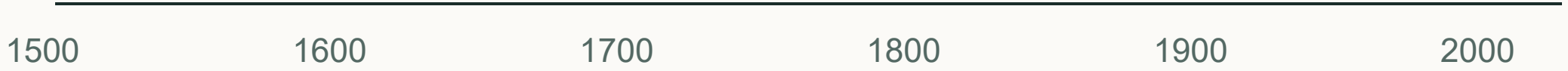
Synthetic teaching data / schematic example; not empirical measurements.

Reference chronologies and calendar anchoring

Begin with securely dated material, such as living trees.



Verify overlaps and extend the record with older wood.



An unanchored sequence is a floating chronology.

Crossdating exercises

A: a complete sequence

Match 30 rings of A to reference M, 1950–1999.

B: a sequence with a shift

Match 31 observed rings and locate the inconsistency.

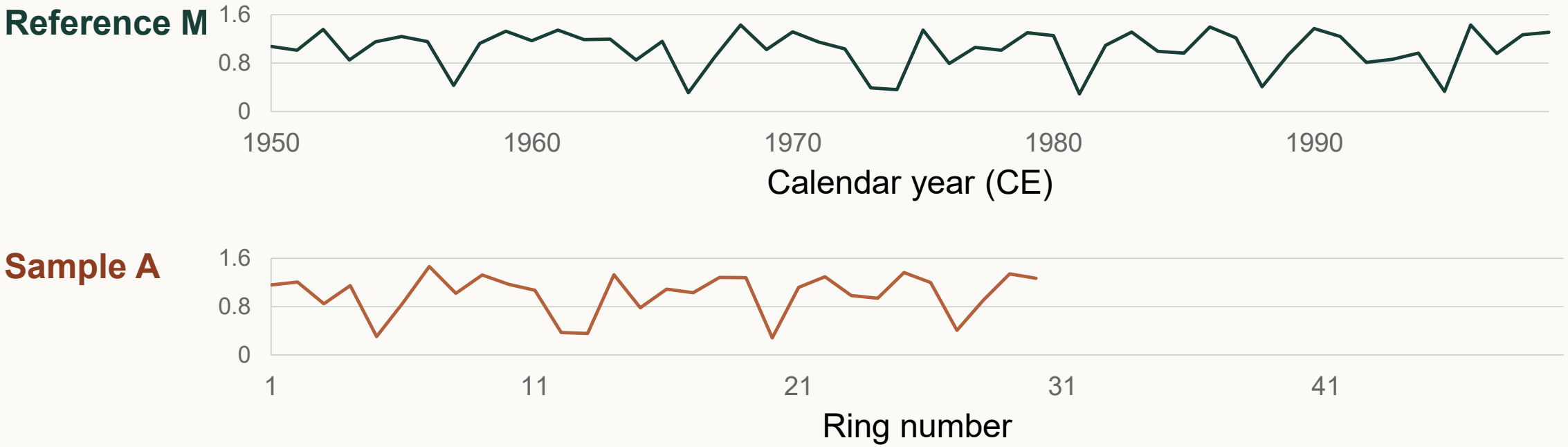
Teaching conditions

All data are synthetic. These short sequences practise a procedure, not real-world dating criteria.

Exercise A: locating 30 rings

Propose the first and last ring years.

Identify at least three matching features and compare another placement.

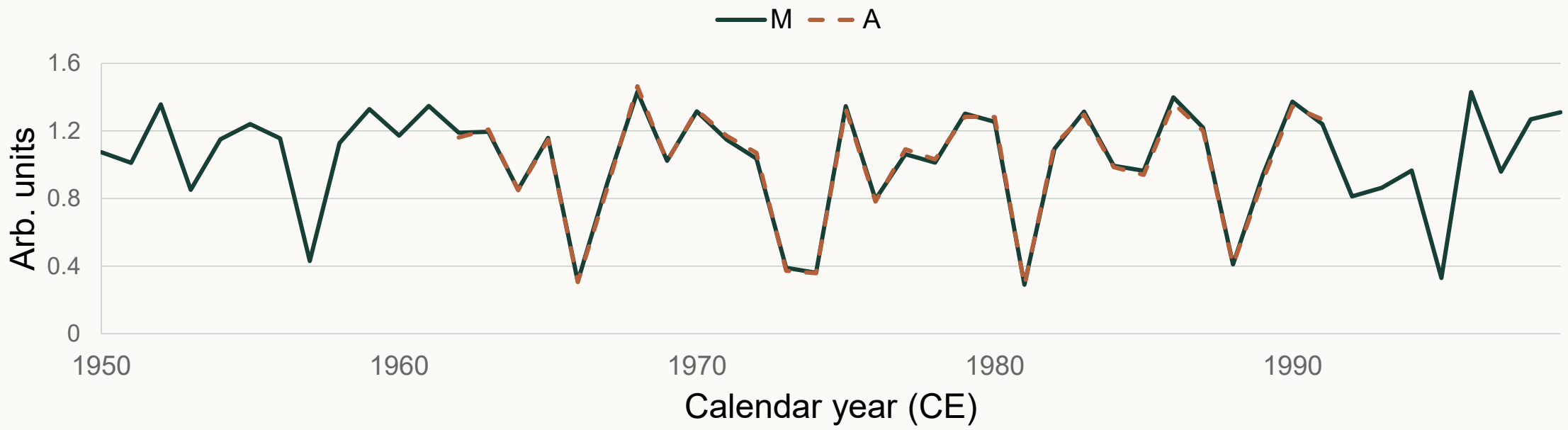


Synthetic teaching data / schematic example; not empirical measurements.

Exercise A: the intended match

A: 1962–1991, 30 rings

Several features align at matching intervals.



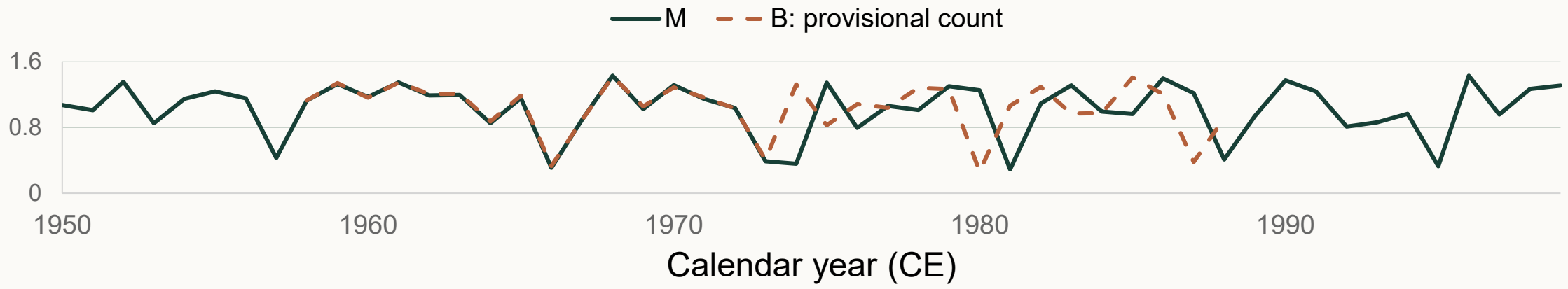
Synthetic teaching data / schematic example; not empirical measurements.

A one-year shift within a sequence

The first part may match while the later part diverges.

Investigate missed boundaries, discontinuous rings or false rings.

Do not insert or delete values merely to improve a match.



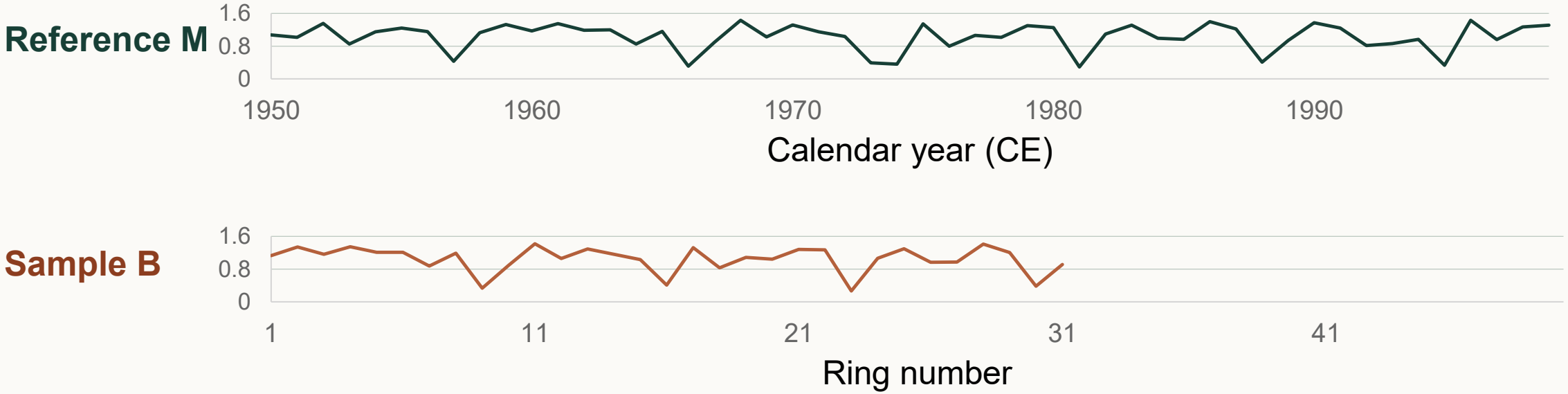
Synthetic teaching data / schematic example; not empirical measurements.

Exercise B: locating the shift

Compare segments of the 31 observed rings.

Can one constant shift align both early and late segments?

Record where to re-examine boundaries and possible causes.



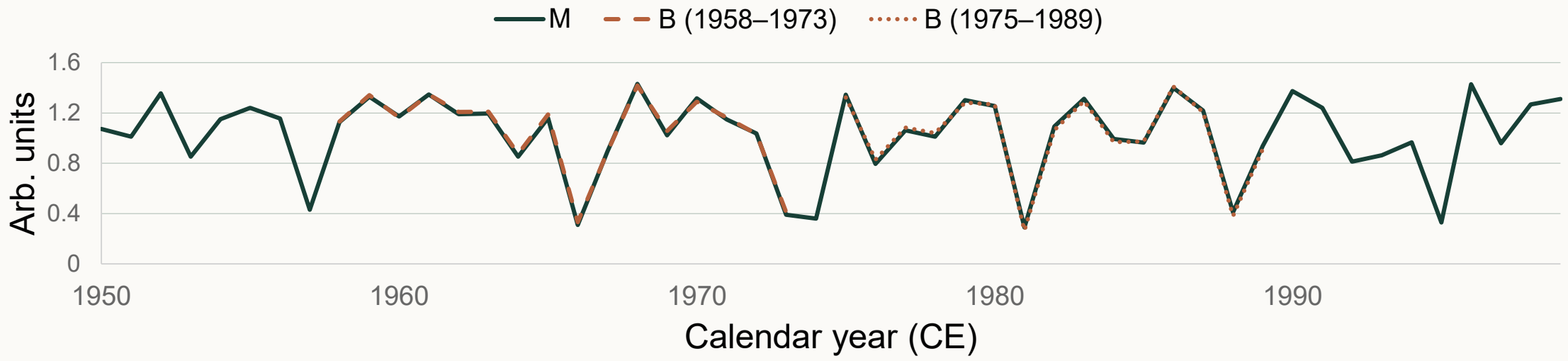
Synthetic teaching data / schematic example; not empirical measurements.

Exercise B: no 1974 increment observed

B spans 1958–1989: 31 observed rings across 32 calendar years.

Reserve the 1974 position between 1973 and 1975.

Keep it missing; do not invent an observed width of zero.



Synthetic teaching data / schematic example; not empirical measurements.

Samples used to verify a match

Another radius of the same tree

Investigate local absences and boundary-reading errors.

A different tree

Check whether the pattern is reproduced.

A reference excluding the unknown

Do not use agreement with a reference already containing that sample as independent proof.

Visual and statistical checks

Correlation helps compare candidate placements.

Its interpretation depends on overlap, preprocessing and search range.

One high coefficient proves neither the date nor its cause.

Candidate start	r (synthetic)	Overlap
1962	0.998	30
1969	0.614	30
1955	0.522	30

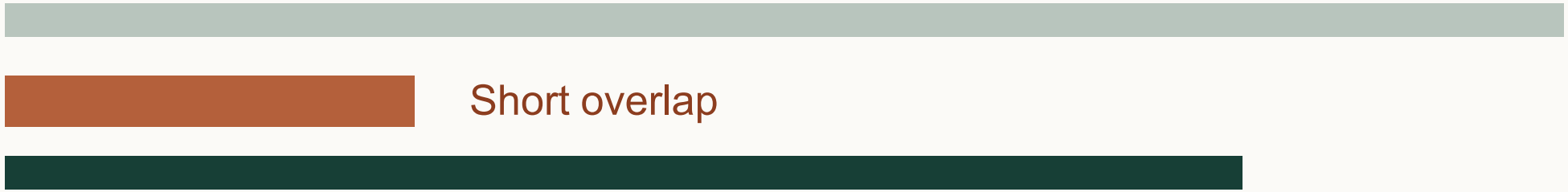
All 21 starts use 30 rings. No p-value or acceptance threshold is implied.

Overlap length and available evidence

Short overlaps provide fewer ways to reject chance matches.

Required length depends on pattern, species, region and other evidence.

A 30-ring exercise does not establish a 30-ring dating rule.



When a sample does not match

Re-examine the evidence

Check surfaces, images, boundaries, measurements and transcription.

Assess reference suitability

Check species, region, period and whether shared variation is expected.

Retain an unresolved result

Do not force calendar dates when evidence is insufficient.

Records supporting a dating decision

01 Sample ID, sampling position and observed boundaries

02 Reference and version, candidate dates and overlap

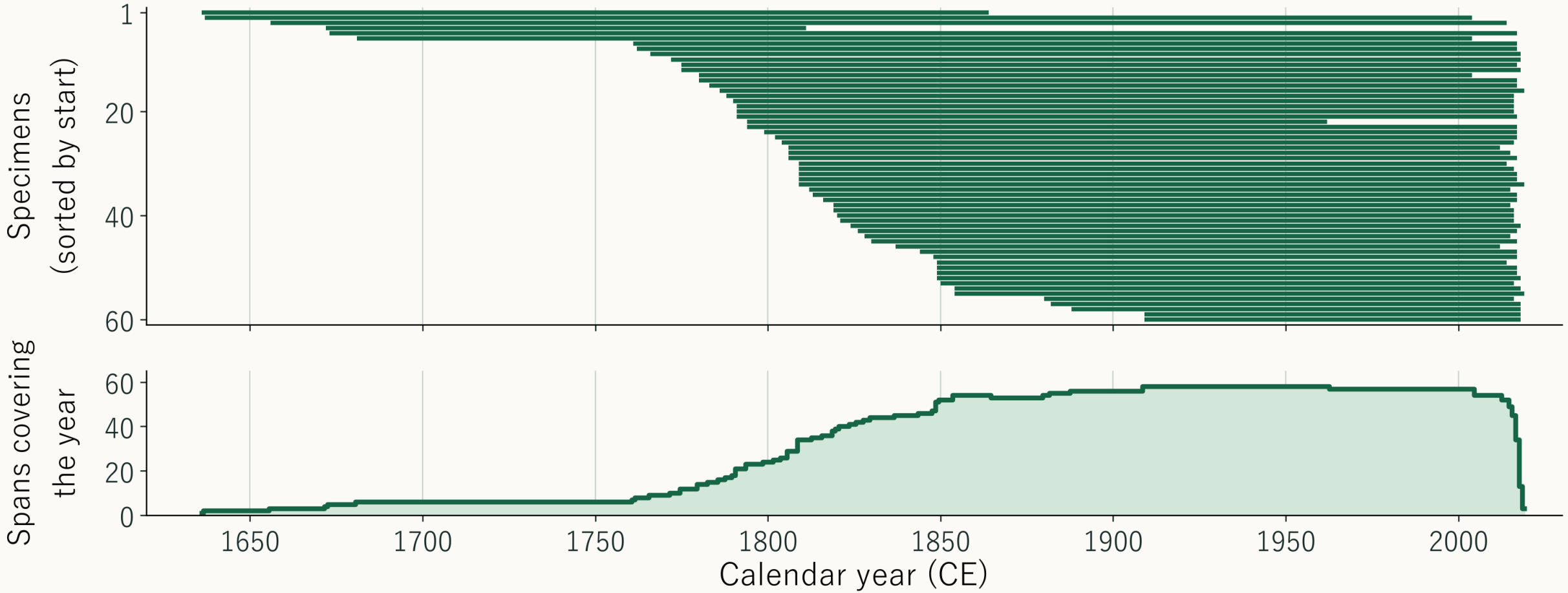
03 Matching and mismatching features, with rechecks

04 Handling of ring anomalies, analyst and date

Yoshino cedar: dates covered by specimens

Open paper

Published start and end years for 60 specimens



Count: specimens with a reported span covering each year.

Coverage does not verify discontinuous rings or valid measurements.

Market-sourced timber; stands and felling cohorts are unknown.

Source: Yonenobu et al. (2026), Supplementary Table S2 | doi.org/10.1016/j.dendro.2026.126572

Summary

Shared variation supports matching

Growth also contains individual and disturbance effects.

Test ring dates with multiple features

Observed ring counts can differ from the calendar span.

Replication and independent checks matter

Investigate mismatches as well as apparent agreement.

Resources and further practice

Instructional crossdating exercises	University of Arizona LTRR
Making skeleton plots	University of Arizona LTRR
The dated master chronology	University of Arizona LTRR
Finding the dates	University of Arizona LTRR
Discontinuous and false rings	University of Arizona LTRR
International Tree-Ring Data Bank	NOAA NCEI

Exit question

1. How can a discontinuous ring affect a count?

2. Is agreement with a mean containing the unknown an independent test?

Write one sentence explaining each answer.

Next: sampling, measurement and ring-width series

Which sample and measurement radius should be used?

Measure widths, preserve records and verify dating.

Preparation: BG01 mathematics/statistics and glossary T32–T38.